

# QUANTITATIVE TRADING WITH MACHINE LEARNING

Paper-inspired forecasting of Volkswagen returns using market and supply-chain signals

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# Project at a glance

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## RESEARCH QUESTION

Can historical VW returns and related-company signals help forecast future returns?

## THREE EXPERIMENTS

1. Understand the reference paper
2. Reproduce the paper-style pipeline
3. Extend the models and compare trading signals

## MAIN TAKEAWAY

The paper-style experiment gave weak aggregate results. The later experiment showed a more promising backtest, but needs stricter out-of-sample validation.

**Important distinction: a better backtest is not automatically proof of better forecasting. The two experiments use different datasets/features and evaluation protocols, so their scores should not be compared as if they were one controlled experiment.**

# Reference paper: what we took from it

## Quantitative Trading with Machine Learning

Stanford CS229 project • René M. Glawion • Spring 2021

Goal: predict future Volkswagen returns using market and supply-chain company features.

Feature space: hundreds of return-related signals from a network of companies.

Models: Elastic Net, Decision Tree, XGBoost and LightGBM.

Evaluation: rolling 5-year train, 1-year validation and 1-month test; repeat monthly.

Horizons: 1, 5, 10, 15, 20 and 25 trading days.

### PAPER LINKS

Report:

<https://cs229.stanford.edu/proj2021spr/report2/81953230.pdf>

Poster:

<https://cs229.stanford.edu/proj2021spr/poster/81953230.pdf>

### REPLICATION REPOSITORY

[https://github.com/rglawion/cs229\\_project\\_report](https://github.com/rglawion/cs229_project_report)

The aim was to learn from the methodology—not assume the same results transfer to a different data pipeline.

# What the reference paper reported

| Paper component   | Reported finding                                              |
|-------------------|---------------------------------------------------------------|
| Model selection   | Elastic Net performed strongly in the paper analysis          |
| Selected strategy | 20-day strategy reported as strongest in the paper's backtest |
| Strategy return   | +21.49% vs +7.98% buy-and-hold (paper-reported)               |
| Forecast errors   | Poster: 1-day RMSE $\approx$ 0.0174; MAE $\approx$ 0.0137     |

**Paper lesson: a sophisticated model is not enough. Time-aware validation and a clearly defined benchmark are central to credible quantitative research.**

## COMPARISON CAVEAT

These are figures reported by the reference paper/poster. They are not directly comparable with our experiments unless dates, target construction, costs and evaluation protocol match.

## WHY REPRODUCE?

To test which findings are robust when we implement the workflow ourselves.

# Our project: old method vs new method

| Dimension     | Old method: paper-style reproduction                      | New method: extended experiment                                            |
|---------------|-----------------------------------------------------------|----------------------------------------------------------------------------|
| Purpose       | Recreate paper's rolling forecasting/selection logic      | Test alternative regressors and a simpler strategy experiment              |
| Data/features | VW + supply-chain proxies; 276 features in old-method PDF | databaseFrankfurtComplete.csv; VW returns + engineered momentum/volatility |
| Models        | Elastic Net, Decision Tree, XGBoost, LightGBM             | Ridge, Elastic Net, Extra Trees                                            |
| Evaluation    | 5y train / 1y validation / 1m test; 55 rolling windows    | Chronological 60% fit / 40% evaluation split                               |
| Main outcome  | Weak aggregate out-of-sample signal                       | More promising strategy backtest, but MAE did not beat zero baseline       |

These are two distinct experiments—not a single experiment with only one model changed.

# Old method: dataset and feature engineering

## TARGET

VWAGY / Volkswagen AG daily prices  
Yahoo Finance • 2015–2025 (as  
documented in old-method PDF)

## PROXIES

9 public related-company proxies:  
Continental, Infineon, HELLA, BASF,  
BorgWarner, Magna, Lear, Aptiv and TSMC.

## FEATURE COUNT

276 total  
24 VWAGY technical features + 252  
proxy-return lags ( $9 \times 28$ )

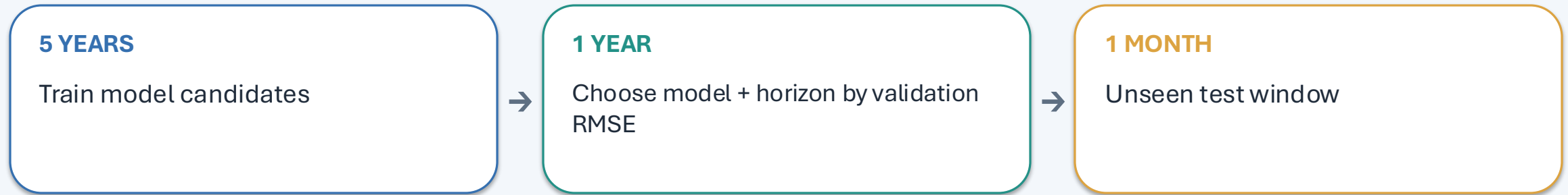
Technical features: 3-, 5-, 10-, 20-, 30- and 60-day returns, rolling volatility, absolute returns and momentum.

Supply-chain proxies: each proxy's historical daily returns lagged from  $t-1$  through  $t-28$ .

Prediction horizons: 1, 5, 10, 15, 20 and 25 trading days.

The proxy set is a practical public-data approximation, not a verified complete supplier list.

# Old method: leakage-aware rolling evaluation



Each window: evaluate 4 models × 6 horizons on validation RMSE.

Select configuration using validation only.

Refit on train + validation; predict the next month.

No random shuffling; chronological split.

Future prices used only as labels.

Test month not used for model selection.

**55 rolling windows were documented in the old-method PDF.**

# Old method: results and interpretation

| Metric           | Old-method result | Interpretation                      |
|------------------|-------------------|-------------------------------------|
| OOS windows      | 55                | Monthly rolling tests               |
| OOS observations | 1,150             | Unseen predictions                  |
| RMSE             | 0.02263           | Aggregate error                     |
| MAE              | 0.01684           | Aggregate absolute error            |
| R <sup>2</sup>   | -0.07275          | Below mean-return baseline          |
| Correlation      | -0.03228          | Little aggregate linear association |
| Positive windows | 8                 | Signal intermittent                 |

## TRADING BACKTEST

Best threshold by Sharpe: 0.50%

ML total return: -13.05%

Buy-and-hold: -70.39%

ML Sharpe: -0.177 vs -0.675

ML max drawdown: -28.63% vs -77.52%

**The ML strategy lost money in aggregate, but had a less severe drawdown than buy-and-hold in this test.**

**Conclusion: weak aggregate predictive performance is a legitimate negative research result; positive individual windows were not stable.**

# Why we added a second experiment

The paper-style experiment did not produce a stable aggregate out-of-sample signal.

We tested whether simpler regularized regression models behaved differently on the supplied Frankfurt dataset.

We added Ridge Regression and compared it with Elastic Net and Extra Trees.

We used the closing-price series to define a forward-return target and engineered additional momentum/volatility features.

The new experiment was treated as a separate model-comparison exercise—not a like-for-like replacement of the old result.

## RIDGE REGRESSION

Objective: minimise squared error plus an L2 penalty on coefficients.

Why try it? Correlated return features can make ordinary estimates unstable.

## HYPOTHESIS

Regularisation may improve generalisation—but must be verified on untouched data.

# New method: data, target and pipeline

## DATA

databaseFrankfurtComplete.csv  
~3,809 rows × 1,292 columns  
2005–2019; ticker VOW.F

## TARGET

Forward return:  
 $y(t) = \text{Close}(t+h) / \text{Close}(t) - 1$   
Example:  $h = 20$  observations

## FEATURES

Historical VW returns + permitted numeric features; added 5/10/20/60-observation momentum and volatility.

## Pipeline

**1**

Load + inspect CSV

**2**

Sort by date

**3**

Build features/target

**4**

Chronological 60:40 split

**5**

Impute missing values

**6**

Scale linear-model inputs

**7**

Fit + predict

**8**

Evaluate forecast + strategy

**Feature availability and leakage checks must be verified before treating the backtest as fully out of sample.**

# New method: models and trading logic

## RIDGE

Linear regression + L2 regularisation  
Alpha = 10.0 in reported run

## ELASTIC NET

L1 + L2 regularisation  
Alpha = 0.01; l1\_ratio = 0.2

## EXTRA TREES

Ensemble of randomized trees  
250 trees; min leaf = 15

## Trading signal

Prediction above positive threshold → LONG.  
Prediction below negative threshold → SHORT (long-short mode).  
Otherwise → NEUTRAL; include turnover cost.

## Evaluation

MAE vs zero-return baseline MAE.  
Directional accuracy and selected threshold.  
Compounded return, Sharpe, max drawdown, invested percentage.

**The reported threshold was selected using fit-period predictions; this can make strategy performance optimistic.**

# New method: reported 60:40 results

| Model       | Test MAE | Zero baseline MAE | MAE improvement | Strategy return | Buy & hold | Sharpe |
|-------------|----------|-------------------|-----------------|-----------------|------------|--------|
| Ridge       | 0.0683   | 0.0671            | -1.79%          | +81.71%         | +10.71%    | 0.500  |
| Extra Trees | 0.0736   | 0.0671            | -9.73%          | +47.97%         | +10.71%    | 0.382  |
| Elastic Net | 0.0684   | 0.0671            | -1.87%          | +43.48%         | +10.71%    | 0.357  |

## WHAT LOOKS BETTER

Ridge had the highest reported strategy return and Sharpe in this run.

## WHAT DOES NOT

All three models had worse MAE than the zero-prediction baseline.

Reported notebook outputs; not independently audited performance.

# Old vs new: how to interpret the results

| Question                  | Old paper-style method                                        | New 60:40 experiment                                            |
|---------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| Predictive signal         | Negative aggregate OOS $R^2$ ; near-zero negative correlation | MAE worse than zero baseline for all tested models              |
| Trading outcome           | Negative ML return; smaller drawdown than buy-and-hold        | Positive reported strategy returns; Ridge strongest in this run |
| Validation strength       | Rolling train/validation/test windows                         | Single chronological split; thresholds tuned on fit predictions |
| Can we say new is better? | Not directly comparable                                       | Promising hypothesis, not confirmed generalisation              |

**The results differ for multiple reasons: data period, feature construction, target definition, model set, validation method and trading rules. A controlled comparison would hold these constant and change only the model.**

# Limitations and next steps

## LIMITATIONS

- Financial returns are noisy and regime-dependent.
- Old and new methods use different datasets and protocols.
- Threshold selection on fitting predictions risks optimism.
- Feature availability at trade time needs verification.
- Costs, slippage and execution are simplified.
- One 60:40 split does not establish robustness.

## NEXT STEPS

- Use purged/embargoed time-series splits for overlapping horizons.
- Select models and thresholds on validation data only.
- Freeze choices before an untouched test period.
- Re-run both approaches on the same dates, target and costs.
- Add verified proxies and macro features.
- Report confidence intervals and regime-wise results.

**Research integrity: report negative outcomes as well as positive backtests, and avoid claiming profitability from one experiment.**

# Suggested live demonstration

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## 01 Data

Show date range, ticker and dimensions.

## 02 Features

Show return lags and engineered features.

## 03 Old method

Show rolling windows and aggregate OOS metrics.

## 04 New method

Show 60:40 split and three regressors.

## 05 Results

Compare MAE baseline, return and Sharpe.

## 06 Limitations

Explain threshold selection and comparability.

Keep the notebook ready at relevant cells; do not spend the demo rerunning every experiment.

## Conclusion & references

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**We implemented a paper-inspired forecasting pipeline, observed weak aggregate out-of-sample performance, then explored additional regressors and trading rules. The extension is useful as an experiment, but requires stricter validation before its stronger backtest can be treated as robust evidence.**

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**Thank you • Questions & discussion**